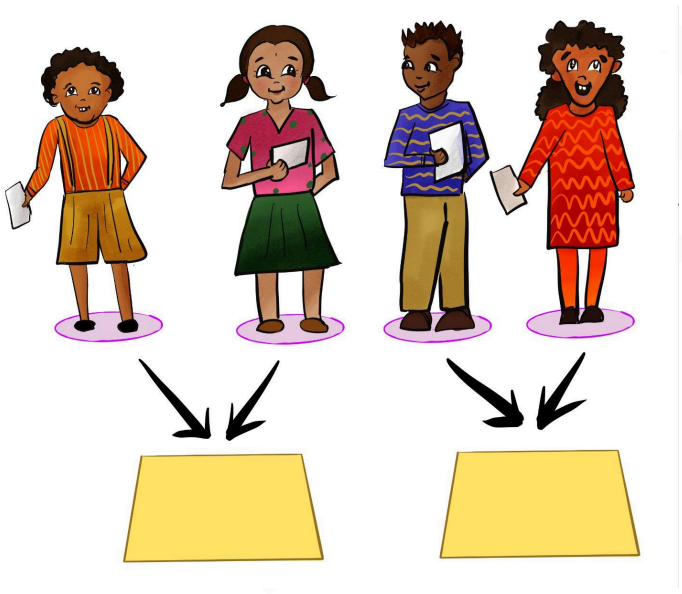
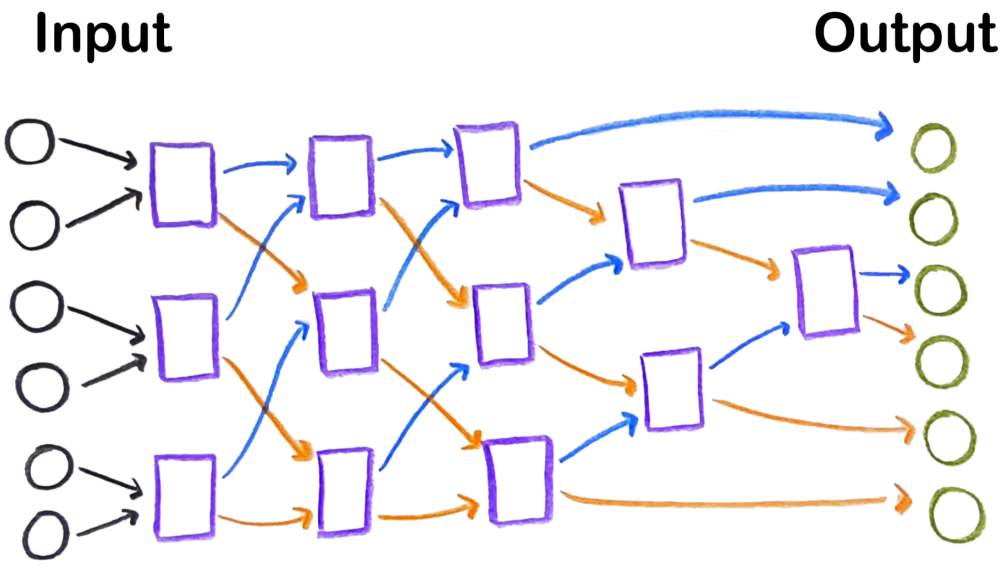
Sorting Networks

Grade 4: Thousands around us

Introduction

Ever wondered how computers sort random things into order so quickly? Books in library, dictionaries, list of students in the classroom, are arranged in an order for convenience. Without sorting, finding information would be much more difficult. This activity uses the network template to be drawn on a floor using chalk (or use flex print) and demonstrates how the numbers can be arrange in an order.





© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Activity	Time	Description
Launch	15-20 mins	Teacher will take the students outdoors and conduct the activity. **Supporting Links:** Network template: https://tinyurl.com/SortingNetworkCT Activity Video: https://tinyurl.com/SortingNetworkVideo Reference: https://tinyurl.com/sortingreference
Trails by Students	10 min	Student's Tryout the activity within a batch of 6. Others can observe the activity and take turns to try out.
Discussions and Explorations	15 min	Attempting the worksheets and the discussion based on the activity

Computational Thinking Components

- *Sorting Algorithm*. The network sorts any given set of numbers in an ordered list. The design of the network and related tasks provides an opportunity to consider the logic and algorithms underlying the activity.
- *Abstraction*. The design of the network is an abstract representation of how data is actually organised in a computer.
- *Decomposition*. The network can start with comparing two numbers and repeating that basic comparison unit; we can sort any set of input numbers.

Activity

This activity uses the network template to be drawn on a floor using chalk (or use flex print) and demonstrates how the numbers can be arranged in order.

- 6 students in the class will take an A4 sheet with numbers 1 to 6 on it.
- Stand on the input circles in any order of the numbers.
- The rectangle box is a comparison box.
- The one with the smaller number will follow the blue, and the other, bigger number will follow the orange.
- At the end of the network, you will see that the numbers are in sorted order.

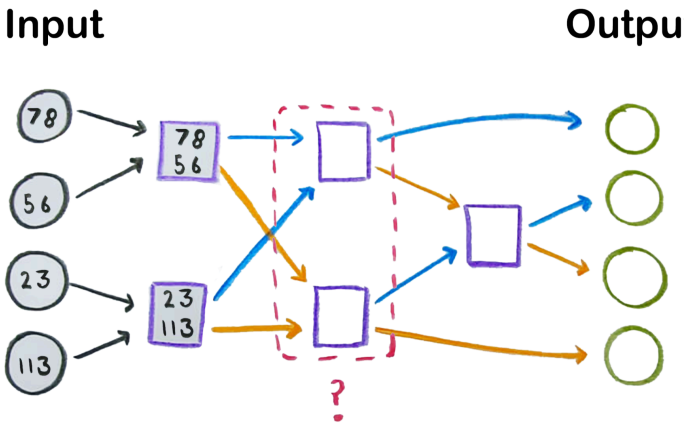
Explorations

Explore the activity with a different order of the numbers in the starting position. Are numbers getting sorted at the end for every input sequence? Instead of numbers 1 to 6, take 2-digit and 3 digits numbers. For example: {23, 56, 78, 113, 378, 667}.

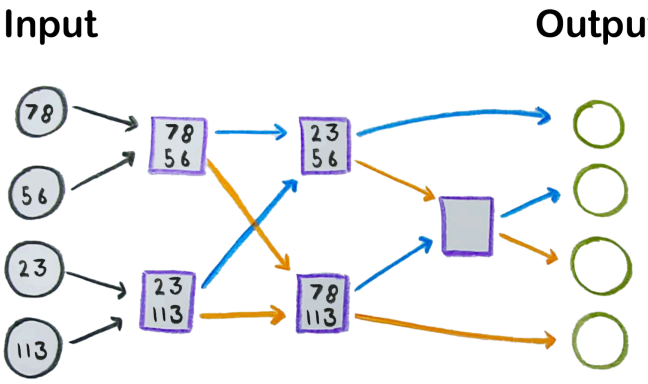
Similarly, the same activity can also be done for sorting students with their names in alphabetical order. Students will ask each other's names in the comparison box. The one with prior in order will go to the left, the other one to the right. See if the names are in order at the end or not at the end. Try the same activity by comparing their height.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

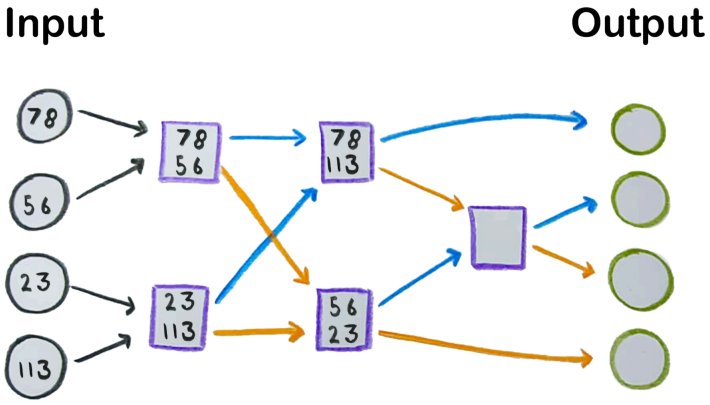
Consider the network to sort any 4 input numbers. The input numbers are {78, 56, 23, 113}. From the input circle position, move the numbers forward to the rectangle box and compare them. The smaller number will follow the blue line and bigger will follow the orange line.



1. **What will be the correct positions of input numbers after the first level of comparison?**



(A)

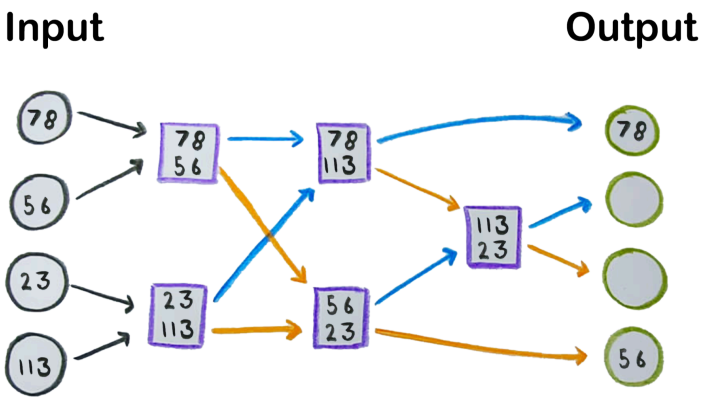


(B)

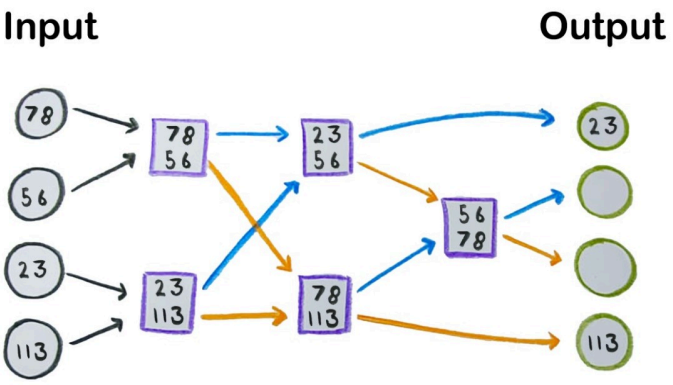
Correct answer: (A)
Explanation: The smaller number move along the blue line.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

2. **What will be the correct positions of numbers after the second level of comparison?**



(A)



(B)

Correct answer: (B)
Explanation: By following the orange line from the upper box, 56 will reach the comparison box. By following the blue line from the lower box, 78, will reach the comparison box.

Now consider the reversal in the rule. Instead of the smaller number, the bigger number from the comparison box will follow the blue arrow. The bigger number will follow the orange arrow.

3. **What will be the output order of the numbers at the end, if the comparison box rule is reversed?**

    a. No difference, output will be the same
    b. Output numbers will be in reverse order - descending
    c. The output depends on the input number order
    d. Can't say anything about the order

Correct answer: b.

Explanation: At every comparison, the bigger number will follow the blue line and move upwards, while the smaller number will follow the orange line and move downwards.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.